MA1513 Exam (Examplify Answers)

AY2022/2023 Semester II

Correct options are highlighted in yellow. Explanatory notes in red.

1. (Linear System) $\left(\begin{array}{ccc|c} a & 1 & 1 & 1 \\ 0 & b & 1 & 1 \\ 0 & 0 & c & 1 \end{array}\right)$

   The above is a row echelon form of an augmented matrix for a linear system with a, b, c being some real numbers. Which of the following statements are true? (Pick all correct options)

   a. This system is consistent for any values of a, b, c.
   b. If the system is consistent, the system has exactly one solution.
   c. It is possible that the system has infinitely many solutions.
   d. The first column of the augmented matrix can be a non-pivot column.
   e. The augmented matrix is full rank for any values of a, b, c.

   a. The value of c in row 3 can be 0, which will make the system inconsistent.
   b. If the system is consistent, then c cannot be zero, which implies a and b are also non-zero. So the first three columns are all pivot columns.
   c. From option b above, we see that it is not possible for the system to have infinitely many solutions.
   d. The value of a is row 1 can be zero (then b and c will also be zero). In this case, the first column is non-pivotal.
   e. Regardless of the values of a, b and c, all the three columns of the row echelon form are non-zero. So the rank is always equal to 3.

2. (Singularity) Given that A and B are non-singular n x n matrices, which of the following statements are always true? (Pick all correct options)

   a. AB is non-singular.
   b. A + B is non-singular.
   c. $A^{-1}B^{-1} = (AB)^{-1}$
   d. The transpose A^T of A is non-singular.
   e. cB is non-singular for any scalar c.

   a. The product of two non-singular matrices is always non-singular.
   b. The sum of two non-singular matrices can be singular or non-singular. For example, if B is -A, then A+B is the zero matrix, which is singular.
   c. Should be $A^{-1}B^{-1} = (BA)^{-1}$
   d. A and A^T have the same determinant, hence share the same singularity.
   e. cB is non-singular all any scalar c except when c = 0.

3. (Determinant) Which of the following determinant has the same value as $\begin{vmatrix} a & b & c \\ 0 & b & c \\ 0 & 0 & c \end{vmatrix}$ for any values of a, b, and c? (Pick all correct options)

   a. $\begin{vmatrix} a & 0 & 0 \\ a & b & 0 \\ a & b & c \end{vmatrix}$
   b. $\begin{vmatrix} 0 & 0 & c \\ 0 & b & c \\ a & b & c \end{vmatrix}$
   c. $\begin{vmatrix} 0 & c & 0 \\ b & c & 0 \\ b & c & a \end{vmatrix}$

d. $\begin{vmatrix} b & a & c \\ 0 & 0 & c \\ b & 0 & c \end{vmatrix}$

e. $\begin{vmatrix} a & 0 & 0 \\ b & b & 0 \\ c & c & c \end{vmatrix}$

$\begin{vmatrix} a & b & c \\ 0 & b & c \\ 0 & 0 & c \end{vmatrix} = abc$. By direct checking, the determinants in options a, d and e are abc; while the determinants in options b and c are -abc.

4. (Span) Which of the following linear spans contain the vector (3,6,9) ? (Pick all correct answers.)
   a. Span{ (9,0,0), (0,6,0), (0,0,3) }
   b. Span{ (9,6,3) }
   c. Span{ (10,20,30) }
   d. Span{ (3,0,9), (0,6,0) }
   e. Span{ (1,1,0), (0,1,1), (1,2,1), (1,0,-1) }

   We check whether (3,6,9) can be expressed as a linear combination of the spanning vectors in each option:
   a: (3,6,9) = 1/3(9,0,0) + (0,6,0) + 3(0,0,3)
   b: (3,6,9) is not scalar multiple of (9,6,3)
   c: (3,6,9) = 3/10(10, 20, 30)
   d: (3,6,9) = (3,0,9) + (0,6,0)
   e: Set up (3,6,9) = a(1,1,0) + b(0,1,1) +c(1,2,1) + d(1,0,-1) and check that there is no solutions for a, b, c, d.

5. (Basis) Which of the following sets of vectors are bases for $\mathbf{R}^4$? (Pick all correct answers)
   a. (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (1, 1, 1, 1)
   b. (1, 1, 1, 1), (1, 0, 1,0), (0,1,0,1), (0,1,1,0)
   c. (1,2,1,1), (1,1,2,1), (1,1,1,2)
   d. (1,2,3,1), (1,4,5,1), (1,6,7, 1), (1,8,9,1)
   e. (1, 1,1,0), (1, 1, 0, 1), (1, 0, 1,1), (0,1,1,1)

   A basis for R^4 must contain 4 linearly independent vectors. We check that options a and e are linearly independent, while options b and d are not linearly independent.

6. (Projection) The projection of the vector (3,3,3) onto the plane with equation x – y + 2z = 0 is given by p = (a, b, c) where a = ___, b = ___, c = ___(FIB)

   Ans: p = (2,4,1)

   To find a basis for the plane, we get two linearly independent vectors that satisfy the equation of the plane: e.g. (1,1,0) and (1,-1,-1) (these vectors are not unique). Then use the basis vectors to form the 3x2 matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 0 & -1 \end{pmatrix}$ with $b = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$. Find the least squares solution of Ax = b, we get $x_0 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$. Then the projection $p = Ax_0 = \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}$.

7. (Coordinate vectors) Let M be a 3x3 non-singular matrix with columns $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$ in that order. Suppose $M\begin{pmatrix}1\\2\\3\end{pmatrix}=\begin{pmatrix}3\\2\\1\end{pmatrix}$. Which of the following statements are true? (Pick all correct answers)

   a. $3\mathbf{c}_1 + 2\mathbf{c}_2 + \mathbf{c}_3$ is the coordinate vector of $\begin{pmatrix}1\\2\\3\end{pmatrix}$ with respect to the basis $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$ of $\mathbf{R}^3$.
   b. $\begin{pmatrix}3\\2\\1\end{pmatrix}$ is the coordinate vector of $\begin{pmatrix}1\\2\\3\end{pmatrix}$ with respect to the basis $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$ of $\mathbf{R}^3$.
   c. $\begin{pmatrix}1\\2\\3\end{pmatrix}$ is the coordinate vector of $\begin{pmatrix}3\\2\\1\end{pmatrix}$ with respect to the basis $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$ of $\mathbf{R}^3$.
   d. $\begin{pmatrix}1\\1\\1\end{pmatrix}$ is the coordinate vector of $\mathbf{c}_1 + \mathbf{c}_2 + \mathbf{c}_3$ with respect to the basis $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$ of $\mathbf{R}^3$.
   e. $\mathbf{c}_1 + \mathbf{c}_2 + \mathbf{c}_3$ is the coordinate vector of $\begin{pmatrix}1\\1\\1\end{pmatrix}$ with respect to the basis $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$ of $\mathbf{R}^3$.

If M = ($\mathbf{c}_1$ $\mathbf{c}_2$ $\mathbf{c}_3$), then the matrix equation $M\begin{pmatrix}1\\2\\3\end{pmatrix}=\begin{pmatrix}3\\2\\1\end{pmatrix}$ can be expressed in vector equation form: $\mathbf{c}_1 + 2\mathbf{c}_2 + 3\mathbf{c}_3 = \begin{pmatrix}3\\2\\1\end{pmatrix}$. By definition of coordinate vectors, this means the statement in option c. Similarly, the statement in option d is correct by definition of coordinate vectors.

8. (Row/Column/Nullspace) Suppose an unknown A is singular and its reduced row echelon form R is known. Which of the following statements are true? (Pick all correct answers)
   a. We can determine the row space of A.
   b. We can determine the column space of A.
   c. We can determine the nullspace of A.
   d. We can determine the matrix A.
   e. We can find a basis for the solution space of Ax = 0.

   a. We can use the non-zero rows of R as the basis for the row space of A, and hence can determine the row space itself.
   b. If A is singular, then R has a zero row. In this case, it is not possible to determine the column space of A.
   c. We can use R to solve the homogeneous system Ax = 0, and hence we can determine the nullspace of A.
   d. It is not possible to determine A from its REF without additional information.
   e. From option c, we can find the solution space, and hence the basis for the solution space.

9. (Linear transformation) Let T: $\mathbf{R}^3 \to \mathbf{R}^3$ be a linear transformation with standard matrix given by A = $\begin{pmatrix}1 & 2 & a\\2 & 0 & b\\1 & -1 & c\end{pmatrix}$. Suppose $T\begin{pmatrix}1\\2\\1\end{pmatrix}=\begin{pmatrix}2\\0\\-1\end{pmatrix}$. Then a = ____, b = ____, c = ____ (FIB)

Ans: a = -3, b = -2, c = 0

This can be solved directly can expanding $\begin{pmatrix}1 & 2 & a\\2 & 0 & b\\1 & -1 & c\end{pmatrix}\begin{pmatrix}1\\2\\1\end{pmatrix}=\begin{pmatrix}2\\0\\-1\end{pmatrix}$.

10. (Diagonalizable)

In the following 3x3 matrices, each * can represent any number.

Which of the following matrices must be diagonalizable? (Pick all correct answers)

a. A = $\begin{pmatrix} 1 & 0 & 0 \\ * & 3 & 0 \\ * & * & 5 \end{pmatrix}$

b. B = $\begin{pmatrix} 6 & * & * \\ 0 & 0 & * \\ 0 & 0 & 6 \end{pmatrix}$

c. C = $\begin{pmatrix} * & 1 & 2 \\ 1 & * & 3 \\ 2 & 3 & * \end{pmatrix}$

d. Matrix D has only 1 eigenvalue and the eigenspace is given by $\mathbf{R}^3$

e. Matrix E can be reduced by Gaussian elimination to a diagonal matrix.

a. A has three distinct eigenvalues 1, 3, 5, hence diagonalizable.

b. B has a repeated eigenvalue 6 with multiplicity 2. It may or may not be diagonalizable.

c. C is a symmetric matrix, hence diagonalizable.

d. Since the eigenspace is the entire $\mathbf{R}^3$, any basis of $\mathbf{R}^3$ will give 3 linearly independent eigenvectors. Hence D is diagonalizable.

e. It is possible to use Gaussian elimination to reduce a non-diagonalizable matrix to a diagonal matrix. E.g. $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \xrightarrow{GE} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

11. (Eigenvector and eigenvalue) Given

A = $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{-1}$ for some real numbers a and b . Which of the following statements are true? (Pick all correct answers)

a. A = $\begin{pmatrix} 7 & 0 & -1 \\ 7 & 0 & 0 \\ 7 & 3 & -1 \end{pmatrix} \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & a \end{pmatrix} \begin{pmatrix} 7 & 0 & -1 \\ 7 & 0 & 0 \\ 7 & 3 & -1 \end{pmatrix}^{-1}$

b. 2A = $\begin{pmatrix} 2 & 2 & 0 \\ 2 & 0 & 0 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} 2a & 0 & 0 \\ 0 & 2a & 0 \\ 0 & 0 & 2b \end{pmatrix} \begin{pmatrix} 2 & 2 & 0 \\ 2 & 0 & 0 \\ 2 & 2 & 2 \end{pmatrix}^{-1}$

c. A^3 = $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^3 \begin{pmatrix} a^3 & 0 & 0 \\ 0 & a^3 & 0 \\ 0 & 0 & b^3 \end{pmatrix} \left(\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{-1} \right)^3$

d. A^{-1} = $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1/a & 0 & 0 \\ 0 & 1/a & 0 \\ 0 & 0 & 1/b \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

e. A = $\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} b & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}^{-1}$

a. The columns in $\begin{pmatrix} 7 & 0 & -1 \\ 7 & 0 & 0 \\ 7 & 3 & -1 \end{pmatrix}$ are scalar multiples of columns in $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$, with columns 2 and 3 swopped. Hence this is a valid diagonalization for A.

b. Matrix 2A has the same eigenvectors as A, with all the eigenvalues doubled: 2a, 2a, 2b. The columns in $\begin{pmatrix} 2 & 2 & 0 \\ 2 & 0 & 0 \\ 2 & 2 & 2 \end{pmatrix}$ are scalar multiples of columns in $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$. Hence this is a valid diagonalization for 2A.

c. The correct diagonalization for A^3 should be

$$A^3 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a^3 & 0 & 0 \\ 0 & a^3 & 0 \\ 0 & 0 & b^3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{-1}$$

d. The correct diagonalization for A^{-1} should be

$$A^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1/a & 0 & 0 \\ 0 & 1/a & 0 \\ 0 & 0 & 1/b \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{-1}$$

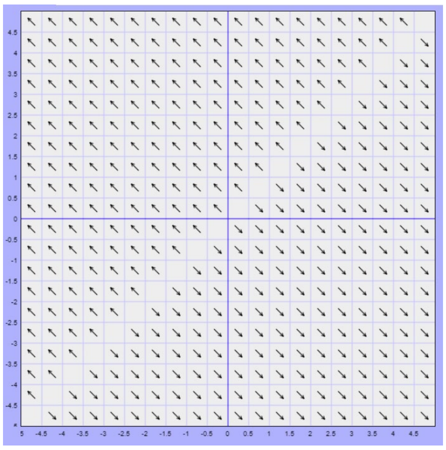
e. The third column of $\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ is not an eigenvector, hence this matrix cannot diagonalize A.

12. (SDE solution) The general solution of an SDE $\begin{cases} y_1' = ay_1 + by_2 \\ y_2' = cy_1 + dy_2 \end{cases}$ is given by $y = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-3t} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t}$. Then a = ____, b = ____, c = ____, d = ____ (FIB)

Ans: a = -5, b = 4, c = -4, d = 5

The underlying matrix A can be found via $A = PDP^{-1}$ where $P = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $D = \begin{pmatrix} -3 & 0 \\ 0 & 3 \end{pmatrix}$.

13. (zero eigenvalue) The direction field of a 2x2 SDE y' = Ay is given below.

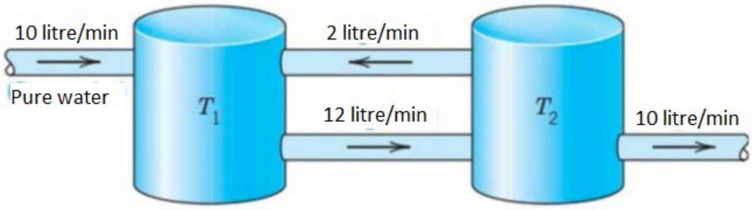


Which of the following statements are true? (Pick all correct answers)

a. A is singular
b. A is not diagonalizable
c. span{(1,1)} is an eigenspaces of A
d. The only equilibrium point of the SDE is at the origin
e. A has a positive eigenvalue

The direction field suggest that phase portrait correspond to the case where A has an eigenvalue 0. Hence A is singular. Since the other eigenvalue is not zero, A has two distinct eigenvalues, hence it is diagonalizable. The eigenvectors corresponding to eigenvalue 0 are along the 45° line, hence the eigenspace is span{(1,1)}. Every point along the 45° line is an equilibrium point. As all the arrows in the direction field moves away from the 45° line, we know the equilibrium points are unstable. Hence the other eigenvalue is positive.

14. (tank problem)
Consider the two tanks shown in the diagram below. Each tank initially holds 100 litre of salt water. The amount of salt at time t (min) in the two tanks are $y_1(t)$ and $y_2(t)$ (kg) respectively.



The SDE that solves for y_1 and y_2 is given by $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$
where a = ___, b = ___, c = ___, d = ___ (FIB) (Give your answers in decimal form with 2 decimal place)

Ans: a = -0.12, b = 0.02, c = 0.12, d = -0.12
The entry a corresponds to the outflow from tank 1, which is given by $-\frac{12}{100}y_1$. So $a = -\frac{12}{100}$.
The entry b corresponds to the inflow into tank 1, which is given by $\frac{2}{100}y_2$. So $b = \frac{2}{100}$.
The entry c corresponds to the inflow into tank 2, which is given by $\frac{12}{100}y_1$. So $c = \frac{12}{100}$.
The entry d corresponds to the outflow from tank 2, which is given by $-\frac{10}{100}y_2 - \frac{2}{100}y_2$. So $d = -\frac{12}{100}$.

15. (Complex eigenvalue) Given the SDE $\begin{cases} y_1' &= 2y_1 &+ y_2 \\ y_2' &= -y_1 &+ 2y_2 \end{cases}$
Which of the following statements are true? (Pick all correct answers)

a. A real solution of the SDE is $e^{2t}\begin{pmatrix} \cos t + \sin t \\ \cos t - \sin t \end{pmatrix}$.
b. $e^{2t}\begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$ is not a real solution of the SDE.
c. The matrix A has an eigenvalue 2+ i with corresponding eigenvector $\begin{pmatrix} 1 \\ -i \end{pmatrix}$.
d. The equilibrium point of the SDE is unstable.
e. The trajectories of the SDE are counter-clockwise.

The underlying matrix $\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$ has eigenvalues $2 \pm i$ with corresponding eigenvectors $\begin{pmatrix} \mp i \\ 1 \end{pmatrix}$. So the fundamental set of real solutions for the SDE is $e^{2t}\begin{pmatrix} \cos t \\ -\sin t \end{pmatrix}$ and $e^{2t}\begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$. Hence $e^{2t}\begin{pmatrix} \cos t + \sin t \\ \cos t - \sin t \end{pmatrix}$ is a real solution. The complex vector $\begin{pmatrix} 1 \\ -i \end{pmatrix} = -i\begin{pmatrix} i \\ 1 \end{pmatrix}$ is a complex scalar multiple of the eigenvector $\begin{pmatrix} i \\ 1 \end{pmatrix}$, corresponding to eigenvalue 2-i, not 2+i. Since the real part of the eigenvalue is positive, the equilibrium point, which is a spiral, is unstable. At the test point (1,0), the tangent vector is $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$, which is pointing downward. So the spiral are moving clockwise.

LQ1

An electrical appliance company produces 3 models (A, B, C) of vacuum cleaners. The amount (in terms of dollar $) of required resources to produce each unit of vacuum cleaners are given below:

	Labour	Material	Miscellaneous
A	30	10	20
B	30	20	30
C	40	20	20

(i) [3 marks] Suppose the company wants to produce 300 units of A, 400 units of B and 500 units of C. Express the budget needed for labour, material and miscellaneous items as a product of a 3×3 matrix with a 3×1 column vector, and write down the three amounts.

(ii) [3 marks] Suppose the budget of the company are: $40,000 on labour, $20,000 on material and $30,000 on other miscellaneous items. Determine the number of units of each model of vacuum cleaners that should be produced if the company wants to use up all of its budget in its production schedule.

(iii) [4 marks] Suppose the company decides to produce one more model D of vacuum cleaners with the same budget as (ii). The labour, material and miscellaneous cost for producing one unit of D are $40, $30 and $30 respectively. Find the largest number of units of D that can be produced if the company wants to use up all of its budget in its production schedule.

(i)

$$\begin{pmatrix} 30 & 30 & 40 \\ 10 & 20 & 20 \\ 20 & 30 & 20 \end{pmatrix} \begin{pmatrix} 300 \\ 400 \\ 500 \end{pmatrix} = \begin{pmatrix} 41000 \\ 21000 \\ 28000 \end{pmatrix}$$

Hence budget for labour is \$41,000, budget for material is \$21,000, and budget for miscellaneous is \$28,000.

(ii)

$$\begin{pmatrix} 30 & 30 & 40 \\ 10 & 20 & 20 \\ 20 & 30 & 20 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 40000 \\ 20000 \\ 30000 \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 30 & 30 & 40 \\ 10 & 20 & 20 \\ 20 & 30 & 20 \end{pmatrix}^{-1} \begin{pmatrix} 40000 \\ 20000 \\ 30000 \end{pmatrix} = \begin{pmatrix} 500 \\ 500 \\ 250 \end{pmatrix}$$

(iii)

$$\begin{pmatrix} 30 & 30 & 40 & 40 \\ 10 & 20 & 20 & 30 \\ 20 & 30 & 20 & 30 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 40000 \\ 20000 \\ 30000 \end{pmatrix}$$

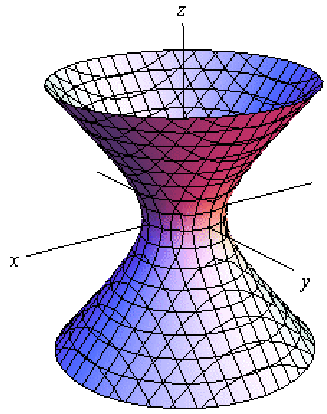
Need to solve by Gaussian elimination:

$$\left(\begin{array}{cccc|c} 30 & 30 & 40 & 40 & 40000 \\ 10 & 20 & 20 & 30 & 20000 \\ 20 & 30 & 20 & 30 & 30000 \end{array}\right) \xrightarrow{G.J.E.} \left(\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 500 \\ 0 & 1 & 0 & 1 & 500 \\ 0 & 0 & 1 & 1 & 250 \end{array}\right)$$

Let $d = t$. Then $c = 250 - t$, $b = 500 - t$, $a = 500 + t$.
The largest number of unit of D is $d = 250$ (so that the number of unit $c = 250 - t$ of C will not become negative).

LQ2

The following surface in the 3D space is known as a hyperboloid. The horizontal cross sections are circles of different radii.



To determine the equation of the surface, which has the form

$$ax^2 + ay^2 + bz^2 = 1 \ (*)$$

for some real numbers a and b, the coordinates of several points at various height on the surface were measured: $(2.65, 1.53, 1), (2.80, 1.62, 2), (3.03, 1.75, 3), (3.33, 1.92, 4), (3.67, 2.12, 5)$.

(i) [3 marks] Set up a 5×2 linear system $\boldsymbol{Ax} = \boldsymbol{B}$ where $\boldsymbol{x} = \begin{pmatrix} a \\ b \end{pmatrix}$, by substituting the given data points into equation $(*)$. Give your answer for $\boldsymbol{A}$ up to 4 decimal places.

(ii) [4 marks] Find the least squares solution of $\boldsymbol{Ax} = \boldsymbol{B}$ and write down the best fit equation for the surface. Give your answer up to 4 decimal places.

(iii) [3 marks] Use the equation in (ii) to estimate the cross sectional radius of the hyperboloid at $z = 10$. Give your answer up to 4 decimal places. Show how you derive the answer.

(i) By substituting the 5 data points into equation $(*)$, we have a linear system:

$$\begin{array}{rcrcl}
(2.65^2+1.53^2)a & + & b & = & 1\\
(2.8^2+1.62^2)a & + & 4b & = & 1\\
(3.03^2+1.75^2)a & + & 9b & = & 1\\
(3.33^2+1.92^2)a & + & 16b & = & 1\\
(3.67^2+2.12^2)a & + & 25b & = & 1
\end{array}$$

So

$$\boldsymbol{A}=\begin{pmatrix}2.65^2+1.53^2 & 1\\ 2.8^2+1.62^2 & 4\\ 3.03^2+1.75^2 & 9\\ 3.33^2+1.92^2 & 16\\ 3.67^2+2.12^2 & 25\end{pmatrix}=\begin{pmatrix}9.3634 & 1\\ 10.4644 & 4\\ 12.2434 & 9\\ 14.7753 & 16\\ 17.9633 & 25\end{pmatrix},\ \boldsymbol{B}=\begin{pmatrix}1\\1\\1\\1\\1\end{pmatrix}$$

(ii) The least squares solution is the solution of $\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}^T\boldsymbol{b}$. So

$$\boldsymbol{x}_0=(\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T\boldsymbol{b}=\begin{pmatrix}0.1109\\-0.0397\end{pmatrix}.$$

So equation of the hyperboloid is

$$0.1109x^2+0.1109y^2-0.0397z^2=1$$

(iii) When $z=10$, the equation gives $0.1109(x^2+y^2)-0.0397(10)^2=1$.
The radius of the cross section is given by $\sqrt{x^2+y^2}=\sqrt{\frac{1+3.97}{0.1109}}=6.6944$.

LQ3

Three telecommunication companies X, Y, Z offering similar mobile plans with some unique features in each plan. According to market research, for each month, there are movements among the clients of the three companies: 2% and 4% of clients of X switch to Y and Z respective; 3% and 3% of clients of Y switch to X and Z respective; 4% and 2% of clients of Z switch to X and Y respective.

(i) [3 marks] Let $\boldsymbol{v}_n = \begin{pmatrix} x_n \\ y_n \\ z_n \end{pmatrix}$ denote the market share of the three companies in the n-th month. Find the matrix $\boldsymbol{A}$ such that $\boldsymbol{v}_{n+1} = \boldsymbol{A}\boldsymbol{v}_n$.

(ii) [4 marks] Write down the n-th power $\boldsymbol{A}^n$ of $\boldsymbol{A}$ in the form $\boldsymbol{P}\boldsymbol{E}\boldsymbol{P}^{-1}$ where $\boldsymbol{E}$ is a diagonal matrix in terms of n.

(iii) [3 marks] Does the market share of the three companies stabilise in the long run? If yes, what is the ratio of the market share of X, Y and Z? Justify your answer.

(iv) [2 marks] How is the long term market share related to the stage matrix $\boldsymbol{A}$?

(a) The system is given by:

$$\begin{aligned} x_{n+1} &= 0.94x_n + 0.03y_n \quad 0.04z_n \\ y_{n+1} &= 0.02x_n + 0.94y_n \quad 0.02z_n \\ z_{n+1} &= 0.04x_n + 0.03y_n \quad 0.94z_n \end{aligned}$$

So the matrix $\boldsymbol{A} = \begin{pmatrix} 0.94 & 0.03 & 0.04 \\ 0.02 & 0.94 & 0.02 \\ 0.04 & 0.03 & 0.94 \end{pmatrix}$.

(b) Using MATLAB, we can write

$$\boldsymbol{A} = \begin{pmatrix} -1 & 1 & 1 \\ 0 & 2/3 & -2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 9/10 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 23/25 \end{pmatrix} \begin{pmatrix} -1 & 1 & 1 \\ 0 & 2/3 & -2 \\ 1 & 1 & 1 \end{pmatrix}^{-1}$$

So

$$\boldsymbol{A}^n = \begin{pmatrix} -1 & 1 & 1 \\ 0 & 2/3 & -2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} (9/10)^n & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & (23/25)^n \end{pmatrix} \begin{pmatrix} -1 & 1 & 1 \\ 0 & 2/3 & -2 \\ 1 & 1 & 1 \end{pmatrix}^{-1}$$

(c) In the long run,

$$\boldsymbol{A}^n = \begin{pmatrix} -1 & 1 & 1 \\ 0 & 2/3 & -2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1 & 1 & 1 \\ 0 & 2/3 & -2 \\ 1 & 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 3/8 & 3/8 & 3/8 \\ 1/4 & 1/4 & 1/4 \\ 3/8 & 3/8 & 3/8 \end{pmatrix}$$

The market share stabilises at the ratio of $3/8 : 1/4 : 3/8$ (or $3 : 2 : 3$).

(d) The vector $\begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix}$ from the market share ratio is an eigenvector of $\boldsymbol{A}$ corresponding to eigenvalue 1.

LQ4

A damped mass-spring system is given by the second order ODE

$$my'' + cy' + ky = 0 \ (*)$$

where m is the mass (in kg) of the ball, k is the spring constant (in Newton/meter) and c is the damping constant(in Newton-seconds/meter). Suppose the ball is 4 kg and the spring constant is 4 Newton/meter.

(i) [2 marks] Suppose the ball returns to equilibrium immediately without oscillation. Find the damping constant c.

(ii) [5 marks] Convert the ODE $(*)$ with the c in (i) to a 2×2 SDE and find a fundamental set of solutions of the SDE.

(iii) [3 marks] Sketch the phase portrait of the SDE in (ii) indicating the directions of the trajectories.

(iv) [3 marks] Explain why the equilibrium point of the SDE that describe any damped mass-spring system cannot be a saddle point regardless of the value of the damping constant c.

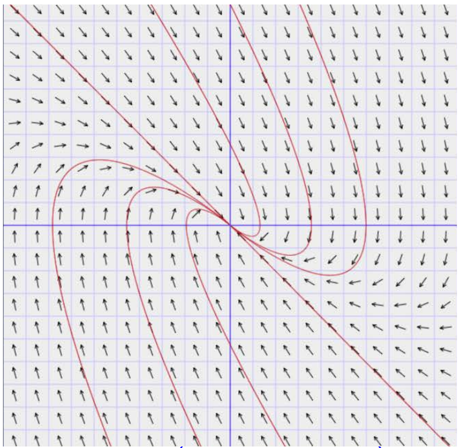
(i) This is critical damping, so $c^2 = 4km = 64$. So $c = 8$.

(ii) The ODE $4y''+8y+4y = 0$ is converted to SDE with underlying matrix $\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix}$

Using MATLAB, We find only one eigenvalue -1 with only one linearly independent eigenvector $\boldsymbol{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. So we need to find a generalised eigenvector $\boldsymbol{u}$ corresponding to $\boldsymbol{v}$ by solving the non-homogeneous system $(\boldsymbol{A} + \boldsymbol{I})\boldsymbol{u} = \boldsymbol{v}$. This gives a general solution $\boldsymbol{u} = \begin{pmatrix} -1-s \\ s \end{pmatrix}$. Taking $s = 0$, we get the generalised eigenvector $\boldsymbol{u} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$.
So a fundamental set of solutions of the SDE:

$$\boldsymbol{v}e^{-t} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-t} \text{ and } (\boldsymbol{v}t + \boldsymbol{u})e^{-t} = \begin{pmatrix} -t-1 \\ t \end{pmatrix} e^{-t}.$$

(iii) Phase portrait



(iv) The underlying matrix $\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ -c/m & -k/m \end{pmatrix}$ has characteristic polynomial $x^2 + (c/m)x + k/m$, where k/m is the product of the two eigenvalues of $\boldsymbol{A}$. For saddle points, the two eigenvalues are of opposite sign, implying k/m is negative. But the spring constant k and the mass of the ball m are both positive quantities. So it is not possible for the equilibrium point to be saddle point.